

Code Pretraining Improves Entity Tracking Abilities of Language Models

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Abstract

Recent work has provided indirect evidence that pretraining language models on code improves the ability of models to track state changes of discourse entities expressed in natural language. In this work, we systematically test this claim by comparing pairs of language models on their entity tracking performance. Critically, the pairs consist of base models and models trained on top of these base models with additional code data. We extend this analysis to additionally examine the effect of math training, another highly structured data type, and alignment tuning, an important step for enhancing the usability of models. We find clear evidence that models additionally trained on large amounts of code outperform the base models. On the other hand, we find no consistent benefit of additional math training or alignment tuning across various model families.

1 Introduction

Entity tracking, the capacity to track how properties of discourse entities and their relationships change as a discourse unfolds, is an important ability for understanding longer contexts as well as other critical capabilities such as planning. For example, to successfully parse the following recipe, an agent needs to track what happens to the different entities, such as ingredients.

- (1) Put the eggs, sugar, flour, and baking powder in a bowl and mix to form a light batter. Make sure that the final batter does not contain any lumps of flour or sugar.

Kim & Schuster (2023) showed that several Transformer-based large language models (LLMs), such as GPT-3.5, exhibit a non-trivial entity tracking capacity. At the same time, they found that similar models, such as GPT-3, seem to lack this ability. Based on the limited information available about the differences between the GPT-3 and GPT-3.5 models, Kim & Schuster (2023) hypothesized that pretraining on large amounts of code imbues LLMs with entity tracking abilities. However, due to the opacity of training data specifications of these models, it remains unclear whether code pretraining indeed is the critical difference. In this work, we re-evaluate this claim with open-source LLMs for which more information about the pretraining process is available. We extend our analysis to the effect of math and instruction tuning as well as code. Upon comparing pairs of base models and models additionally trained on code, math, or alignment tuned,¹ we find a clear benefit of code training but no consistent benefit of math training or alignment tuning.

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¹We use the term alignment tuning to refer to various methods of making language models more useful for interactive settings, including supervised instruction finetuning (SFT) (Wei et al., 2022a), reinforcement learning from human feedback (RLHF) (Ouyang et al., 2022), and direct preference optimization (DPO) (Rafailov et al., 2023).

Exp	Model	Size	Additional Training			Base Model
			+Code	+Math	+Instruct/Chat	
Exp 1 (code)	Llama 2	7B–70B				-
	Code Llama	7B–70B	✓			Llama 2
	DeepSeek	7B				-
	DeepSeek-Coder	7B	✓			DeepSeek
	Gemma	8B				-
	CodeGemma	8B	✓			Gemma
Exp 2 (math)	Llama	7B				-
	FLoat	7B		✓ (instruct)		Llama
	Mistral	7B				-
	OpenMathMistral	7B		✓ (instruct)		Mistral
	DeepSeek-Coder	7B	✓			DeepSeek
	DeepSeek-Math	7B	✓	✓		DeepSeek-Coder
	Code Llama	7B, 34B	✓			Llama 2
	Llemma	7B, 34B	✓	✓		Code Llama
Exp 3 (alignment)	Llama 2	7B–70B				-
	Llama 2-Chat	7B–70B			✓	Llama 2
	Code Llama	7B–70B	✓			Llama 2
	Code Llama-Instruct	7B–70B	✓		✓	Code Llama 2
	Gemma	8B				-
	Gemma-Instruct	8B			✓	Gemma
	CodeGemma	8B	✓			Gemma
	CodeGemma-Instruct	8B	✓		✓	CodeGemma
	DeepSeek	7B				-
	DeepSeek-Chat	7B			✓	DeepSeek
	DeepSeek-Coder	7B	✓			DeepSeek
	DeepSeek-Coder-Instruct	7B	✓		✓	DeepSeek-Coder

Table 1: Summary of the models compared and their pretraining data composition.

2 Related work

Including code in the pretraining data mixture, even for models not explicitly specialized for code, has become increasingly customary in LLM training (Chowdhery et al., 2023; Touvron et al., 2023b; Gemini Team et al., 2024; Groeneveld et al., 2024, *i.a.*). In addition to serving the popular use case of LLMs in code completion and generation (Chen et al., 2021), adding code to the pretraining data mixture has been claimed to improve general reasoning capacities of LLMs (Fu et al., 2022; Ma et al., 2024; Yang et al., 2024). Kim & Schuster (2023) hypothesized that a concrete capacity that can benefit from code is entity tracking: converging evidence towards this claim is contributed by observations from Madaan et al. (2022) (code pretrained models like Codex perform better than models primarily trained on language data on ProPara (Dalvi et al., 2019)), Sap et al. (2022) (GPT-3.5 performs better on object tracking than GPT-3), and Muennighoff et al. (2023) (adding code to the pretraining data improves performance the on bAbI tasks (Weston et al., 2016)). Furthermore, Prakash et al. (2024) observed that a base model finetuned on arithmetic tasks improved performance on a simplified version of the entity tracking task by Kim & Schuster (2023), suggesting that structured data in general beyond code may contribute to the development of an entity tracking capacity in language models.

3 Experiments

We aim to systematically test the hypothesis that code pretraining leads to better entity tracking put forward by Kim & Schuster (2023), through a series of experiments comparing base models and models continued to be trained on code on top of the base models. We additionally test the hypothesis that pretraining on math, another type of structured data, leads to better entity tracking performance through similar comparisons.

2-shot prompt

Given the description after "Description:", write a true statement about all boxes and their contents to the description after "Statement:".

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map.

Statement: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map.

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. Remove the car from Box 0. Remove the paper and the string from Box 3. Put the plane into Box 0. Move the map from Box 6 to Box 2. Remove the bill from Box 4. Put the coat into Box 3.

Statement: Box 0 contains the plane, Box 1 contains the cross, Box 2 contains the bag and the machine and the map, Box 3 contains the coat, Box 4 contains nothing, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle.

Description: {description}

Statement: Box 0 contains

Table 2: Prompts with 2-shot in-context demonstrations.

3.1 Models

We selected model pairs that have been reported to vary only in terms of their pretraining data. For testing the code hypothesis, we compared the following pairs of models: (Llama 2, Code Llama), (DeepSeek, DeepSeek-Coder), and (Gemma, CodeGemma), where the second model in each pair is obtained by continuing to train the first model on additional code data. We tested 7B, 13B, and 70B models in the Llama 2 series. For testing the math hypothesis, we compared the following four pairs of models: (Code Llama, Llemma), (DeepSeek-Coder, DeepSeek-Math), (Llama, FLoat), and (Mistral, OpenMathMistral). Again, the second model in each pair is obtained by training the first model on additional math data. For alignment tuning, we compared (Llama 2, Llama 2-chat), (Code Llama, Code Llama-Instruct), (Gemma, Gemma-Instruct), (CodeGemma, CodeGemma-Instruct), (DeepSeek, DeepSeek-Chat), and (DeepSeek-Coder, DeepSeek-Coder-Instruct). These comparisons are summarized in Table 1. See Appendix A.1, Table 4 for more details about the models.

3.2 Evaluation setup

We adopted the boxes task (the “base” version) from Kim & Schuster (2023) for testing the models’ entity tracking capacity. In this task, the input to the LLM is a textual description of the contents of seven boxes followed by 1–12 descriptions of operations that change the contents of the individual boxes. In response to this input, the LLM is prompted to state the contents of each box according to the initial description and the state-changing operations. We used the same prompt and 2-shot in-context learning examples as Kim & Schuster (2023) (see Table 2 for an example). We used a slightly different prompt format for chat-optimized models to align the task better to the input format the models were trained on. The inputs to the model are provided as “user” prompts, and the expected model outputs are formatted as “assistant” (see Table 5 in Appendix for an example prompt).

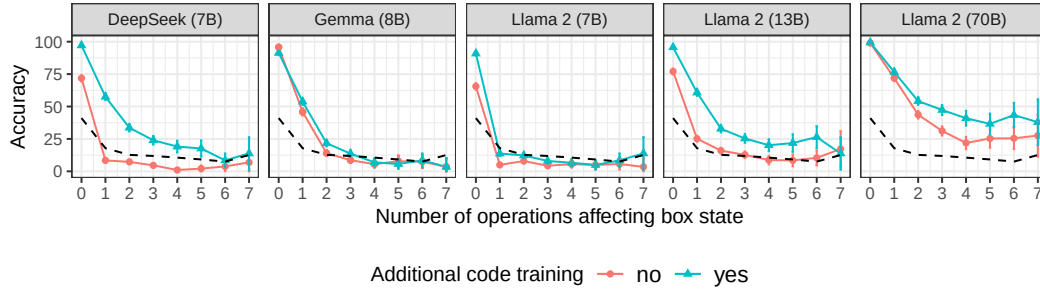


Figure 1: Entity tracking results for DeepSeek, Gemma, and Llama 2 models. Error bars indicate 95% confidence intervals, and the black dashed lines show the performance of the random baseline.

We noticed that smaller models suffered from formatting issues, often deviating from the format specified by the prompt or omitting the contents of some boxes. For this reason, we used regular expression-based constrained decoding using the `outlines` library (Willard & Louf, 2023).²

We report all results divided into the number of operations affecting the target box rather than reporting one aggregate accuracy metric. This is to distinguish trivial cases from cases that actually require tracking state changes—when the number of operations affecting the target box is 0, simply copying from the initial state description yields the correct answer. Furthermore, we compare the model results to the strong random baseline by Kim & Schuster (2023). For this baseline, we randomly sample 0 to 3 objects for each box from the set of objects that have been previously mentioned in a clause with the box in question.

3.3 Experiment 1: Effect of Code

Figure 1 compares the entity tracking performance of base models (red lines) and code models (blue lines) for models from the DeepSeek (DeepSeek-AI et al., 2024; Guo et al., 2024), Gemma (Gemma Team et al., 2024), and Llama 2 families of various sizes (Touvron et al., 2023b; Rozière et al., 2024). In general, we find clear evidence that continued training on large amounts of code improves entity tracking abilities, as can be seen for the Llama 2 13B and 70B models as well as for the DeepSeek models. In these model comparisons, the models trained on code consistently outperformed the base models on the nontrivial cases of entity tracking (number of operations affecting box state ≥ 1). In the case of 13B models, a boost in trivial cases is also observed (number of operations = 0); in 70B models, performance on the trivial cases is already saturated in the base model.

In Llama 2 7B models, the gains through additional code training are relatively minor, with most of the gains deriving from boosts in examples where the number of operations is either 0 or 1. Similarly minor gains were observed in CodeGemma 8B, except that the gains were observed in examples with 1 and 2 operations. For both of these models, we also observed that for number of operations greater than 0 (Llama 2 7B) and 2 (Gemma 8B), neither the base nor the code variants perform better than our random baseline. These results suggest that there is both a possible effect of scale in the effectiveness of code training as observed in the Llama 2 series, and an effect of the amount of additional code training (DeepSeek-Coder: 2T tokens, Code Llama: 500B tokens).

3.4 Experiment 2: Effect of Math

In evaluating the effect of additional math training, we start by revisiting the claim of Prakash et al. (2024) that the FLoat model obtained by finetuning Llama 7B (Touvron et al., 2023a) on arithmetic tasks from Liu & Low (2023) yields superior entity tracking

²<https://github.com/outlines-dev/outlines>

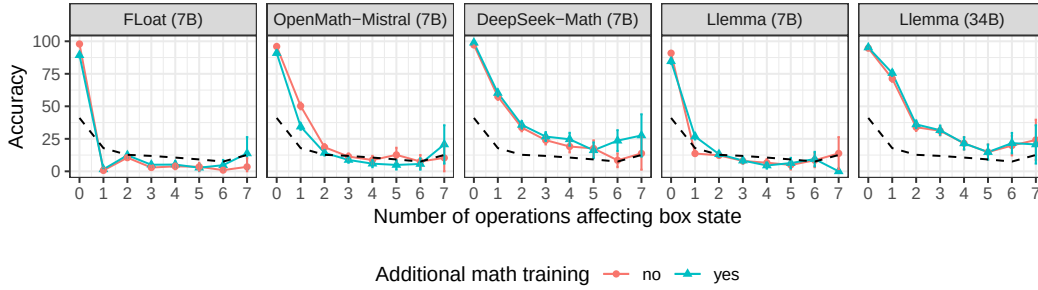


Figure 2: Entity tracking results for models trained with additional math data. See Table 1 for the model names of the base and math models. Error bars indicate 95% confidence intervals, and the black dashed lines show the performance of the random baseline.

performance. As can be seen in Table 3, FLoat did show slightly higher accuracy on the non-trivial tracking cases (number of operations ≥ 1) than the base model, but the gain was marginal.³ Furthermore, neither the base Llama model nor the FLoat model performed better than our random baseline on non-trivial entity tracking examples (Figure 2, far left).

Model	Aggregate	NumOps = 0	NumOps ≥ 1
Llama 7B	28.67	97.93	4.34
FLoat 7B	27.55	89.33	5.85

Table 3: Llama 7B vs. FLoat 7B results.

Following this observation, we compared Mistral and OpenMathMistral models where the latter is a model further trained on OpenMathInstruct-1, a synthetically generated instruction-tuning dataset containing 1.8M unique solutions to math problems sourced from MATH and GSM8K datasets (Toshniwal et al., 2024). As shown in Figure 2, OpenMathMistral only achieved marginal gains over the base Mistral model when there are 7 operations affecting the target box, and in most other cases, the base model consistently outperforms the math-finetuned model. Furthermore, neither model outperformed the random baseline for examples with more than 2 operations affecting the box of interest.

The unclear benefit of additional math training is further corroborated by marginal gains in models trained on math data that are not in “instruct” format like FLoat and OpenMathMistral. Figure 2 shows that DeepSeek-Math (Shao et al., 2024) performed close to the DeepSeek-Coder model for most cases. The gains are even more limited in the comparison between Code Llama vs Llemma (Azerbayev et al., 2024). Llemma 34B outperformed Code Llama 34B by a narrow margin for the non-trivial tracking cases (Llemma: 47.86, Code Llama: 45.46 for examples where the number of operations ≥ 1). These results suggest a limited benefit of additional math pretraining on entity tracking.

3.5 Experiment 3: Effect of Alignment Tuning

Finally, we explore the effect of alignment tuning on entity tracking. For models in the Llama 2 family, alignment tuning the base models led to minor gains (Figure 3, orange vs. green lines in the top row panels), whereas alignment-tuned code models did not consistently lead to gains and sometimes performed worse than the non-alignment-tuned counterparts (orange vs. green lines in the bottom row panels). Nevertheless, the best-performing model was CodeLlama 70B-instruct (64.9 accuracy on 1+ operations), combining code and alignment tuning.

³These numbers are not expected to align with numbers reported in Prakash et al. (2024) because they used a modified version of the original task.

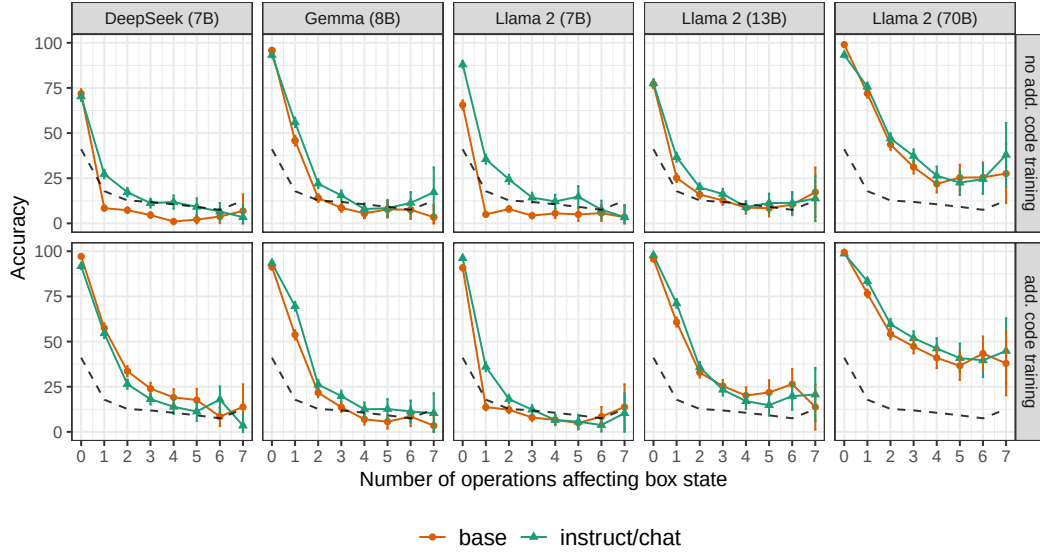


Figure 3: Entity tracking results for alignment-tuned DeepSeek, Gemma and Llama 2 models. The top panels show models without additional code training, whereas the bottom panels show models that have been trained on additional amounts of code before alignment tuning. Error bars indicate 95% confidence intervals, and the black dashed lines show the performance of the random baseline.

The DeepSeek models showed similar trends to the general observation made above: alignment tuning of the base model led to gains, whereas alignment tuning of the code model did not. Gemma 8B and CodeGemma 8B models did benefit from alignment tuning, similarly to Llama 2 7B and CodeLlama 7B models, although the gains were smaller.

Overall, alignment tuning affects base and code models differently, where the gains for base models tend to be greater. The benefit of alignment tuning for base models seems to be inversely correlated with scale: smaller base models benefit more from alignment tuning.

4 Conclusion and Future Work

We explored the effect of code, math, and alignment tuning on LLMs’ capacity to track entities in natural language text. Our main findings are threefold:

1. Additional code training leads to consistent improvements across model families and sizes.
2. Additional math training does not yield consistent improvements, and the performance gains are at best marginal.
3. Alignment tuning leads to different patterns of improvement depending on whether it was applied to base models or code models. Base models consistently benefit from alignment tuning, and smaller models see more improvement. The benefit for code models is more mixed, but the best performance is achieved through combining code and instruction tuning.

Our work thus adds to a growing body of literature that suggests that pretraining on code improves LLM performance on reasoning tasks, including commonsense reasoning (Madaan et al., 2022), chain-of-thought reasoning (Wei et al., 2022b), mathematical problems (Razeghi et al., 2024), and entity tracking tasks (Muennighoff et al., 2023). Why might this be the case? Kim & Schuster (2023) argued that keeping track of the states of variables is important for producing correct code, and hypothesized that this kind of procedural input may provide a

stronger training signal than pure natural language text. We consider investigating how code training imbues models with entity tracking and other reasoning abilities an important direction for future research.

Limitations While the pairs of models we compared are “minimal pairs”, several possible confounds remain in our interpretation. For example, we interpreted parts of the results in Experiment 2 as marginal benefit of math compared to code, but the OpenMathInstruct dataset (1.5 GB) is two orders of magnitude smaller in terms of the number of tokens compared to Code Llama’s code data (500 B tokens), so the size of the additional training data could be a confound. The additional math training data of DeepSeek-Math is more comparable (120B tokens), but we do not have a model that is only continually trained on math data; DeepSeek-Math is trained on both code and math. Furthermore, the math data vary along several other important dimensions: OpenMathInstruct, and FLoat models use synthetic data, whereas others use naturally occurring data. FLoat and OpenMathMistral are tuned on math data in instruction format, whereas the training data of DeepSeek-Math and Llemma are not. Unfortunately, we cannot fully tease apart the effect of the format of the math data (instruct vs. non-instruct) here because the format co-varies with whether code was additionally in the training mixture: all models that were additionally trained with non-instruction-formatted math data were continuously trained from models that were trained on code already. Lastly, for the experiments investigating the effect of alignment tuning, we considered models that were alignment tuned through a range of different methods and types of data, and some of the diverging findings in this experiment may be attributed to these differences. We plan to address these existing limitations through controlled training experiments in future investigations.

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Model	Size	Hugging Face Identifier	Chat format
Llama 2	7B	meta-llama/Llama-2-7b-hf	–
	13B	meta-llama/Llama-2-13b-hf	–
	70B	meta-llama/Llama-2-70b-hf	–
Llama 2-Chat	7B	meta-llama/Llama-2-7b-chat-hf	✓
	13B	meta-llama/Llama-2-13b-chat-hf	✓
	70B	meta-llama/Llama-2-70b-chat-hf	✓
Code Llama	7B	codellama/CodeLlama-7b-hf	–
	13B	codellama/CodeLlama-13b-hf	–
	34B	codellama/CodeLlama-34b-hf	–
	70B	codellama/CodeLlama-70b-hf	–
Code Llama-Instruct	7B	codellama/CodeLlama-7b-Instruct-hf	✗
	13B	codellama/CodeLlama-13b-Instruct-hf	✗
	34B	codellama/CodeLlama-34b-Instruct-hf	✗
	70B	codellama/CodeLlama-70b-Instruct-hf	✗
DeepSeek	7B	deepseek-ai/deepseek-llm-7b-base	–
DeepSeek-Chat	7B	deepseek-ai/deepseek-llm-7b-chat	✓
DeepSeek-Coder	7B	deepseek-ai/deepseek-coder-7b-base-v1.5	–
DeepSeek-Coder-Instruct	7B	deepseek-ai/deepseek-coder-7b-instruct-v1.5	✗
DeepSeek-Math	7B	deepseek-ai/deepseek-math-7b-base	–
Gemma	8B	google/gemma-7b	–
Gemma-Instruct	8B	google/gemma-1.1-7b-it	✗
CodeGemma	8B	google/codegemma-7b	–
CodeGemma-Instruct	8B	google/codegemma-1.1-7b-it	✗
Llemma	7B	EleutherAI/llemma_7b	–
	34B	EleutherAI/llemma_34b	–
Llama	7B	huggyllama/llama-7b	–
FLoat	7B	nikhil07prakash/float-7b	–
Mistral	7B	mistralai/Mistral-7B-v0.1	–
OpenMathMistral	7B	nvidia/OpenMath-Mistral-7B-v0.1	–

Table 4: Details of all the models evaluated in the paper. The rightmost column indicates whether the chat format was used for prompting the model.

A Appendix

A.1 Model Details

We used the transformers library (Wolf et al., 2020) by Hugging Face for all our experiments. Table 4 presents all the models along with their Hugging Face identifiers.

For alignment-tuned models, we experimented with prompting the model with and without chat formatting (see Table 5 and Table 2 for the different formats). Based on results over a held-out development set, we selected the best-performing prompt format. We find that except for Llama 2-Chat models and DeepSeek-Chat, all other alignment-tuned models performed better with the non-chat format.

A.2 Constrained Decoding

```
Statement: Box 0 contains( [a-zA-Z]+)*, Box 1 contains( [a-zA-Z]+)*, Box
2 contains( [a-zA-Z]+)*, Box 3 contains( [a-zA-Z]+)*, Box 4 contains(
[a-zA-Z]+)*, Box 5 contains( [a-zA-Z]+)*, Box 6 contains( [a-zA-Z]+)*.
```

Figure 4: Regular expression used for constrained decoding of entity states.

2-shot prompt

<s>[INST] «SYS» Given the description after "Description:", write a true statement about all boxes and their contents to the description after "Statement:". «/SYS»

Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. [/INST] Statement: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. </s>

<s>[INST] Description: Box 0 contains the car, Box 1 contains the cross, Box 2 contains the bag and the machine, Box 3 contains the paper and the string, Box 4 contains the bill, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle and the map. Remove the car from Box 0. Remove the paper and the string from Box 3. Put the plane into Box 0. Move the map from Box 6 to Box 2. Remove the bill from Box 4. Put the coat into Box 3. [/INST] Statement: Box 0 contains the plane, Box 1 contains the cross, Box 2 contains the bag and the machine and the map, Box 3 contains the coat, Box 4 contains nothing, Box 5 contains the apple and the cash and the glass, Box 6 contains the bottle. </s>

<s>[INST] Description: description

Table 5: Chat-formatted prompt with 2-shot in-context demonstrations.

In our experiments, we found that the models struggled to adhere to the output format specified via the few-shot prompt examples. Luckily, the expected output can be described precisely by the regular expression shown in Figure 4. We used the outlines library (Willard & Louf, 2023) which supports regex-based constrained decoding. We found a significant improvement with constrained decoding. For e.g., the performance of the Llama 2 70B model went up from 54.95 to 62.13 with constrained decoding.

A.3 Detailed Results

Table 6 presents the detailed results of all the models evaluated in this work. The results are categorized by the number of operations affecting the entity of interest.

Model	Performance split by number of operations								
	Overall (5012)	0 (1303)	1 (1410)	2 (1083)	3 (651)	4 (288)	5 (142)	6 (106)	7 (29)
Random	21.08	41.06	17.85	12.70	11.87	10.58	9.16	7.51	12.59
Llama 2-7B	21.31	65.54	4.96	7.94	4.30	5.56	4.93	5.66	3.45
Llama 2-7B Chat	41.28	87.95	35.53	24.38	14.29	12.15	14.79	7.55	3.45
Llama 2-13B	33.28	77.05	25.18	15.97	12.75	8.68	8.45	10.38	17.24
Llama 2-13B Chat	38.05	77.51	36.67	19.94	16.28	9.03	11.27	11.32	13.79
Llama-2 70B	62.13	99.00	71.91	43.67	31.18	21.88	25.35	25.47	27.59
Llama-2 70B Chat	63.43	93.25	75.67	47.00	37.33	26.39	22.54	24.53	37.93
Code Llama 7B	31.94	90.87	13.69	12.28	7.99	6.60	4.93	8.49	13.79
Code Llama 7B Instruct	41.34	96.16	36.03	18.19	12.29	6.60	5.63	3.77	10.34
Code Llama 13B	54.79	95.70	60.78	32.87	25.35	20.14	21.83	26.42	13.79
Code Llama 13B Instruct	58.14	97.77	71.13	35.83	23.35	17.01	14.79	19.81	20.69
Code Llama 34B	58.26	94.70	71.28	33.89	31.18	21.53	14.79	19.81	24.14
Code Llama 34B Instruct	61.47	95.09	77.09	39.98	31.03	22.57	19.01	20.75	20.69
Code Llama 70B	69.77	99.39	76.60	54.20	47.31	40.97	36.62	43.40	37.93
Code Llama 70B Instruct	73.66	98.70	83.40	59.65	51.92	46.18	40.85	39.62	44.83
Llemma 7B	34.12	84.65	26.81	13.20	8.29	4.51	6.34	9.43	0.00
Llemma 34B	60.18	95.24	75.60	36.10	31.64	21.53	14.79	21.70	20.69
DeepSeek 7B	23.46	71.83	8.44	7.29	4.61	1.04	2.11	3.77	6.90
DeepSeek 7B Chat	32.24	70.30	27.30	17.27	11.21	11.81	9.15	6.60	3.45
DeepSeek 7B Coder Base	53.67	97.16	57.52	33.61	23.96	19.10	17.61	8.49	13.79
DeepSeek 7B Coder Instruct	48.76	91.71	54.54	26.41	18.13	13.89	11.27	17.92	3.45
DeepSeek 7B Math Base	56.40	98.93	60.28	35.73	26.73	24.65	16.20	23.58	27.59
Gemma-7B	42.66	95.78	45.82	14.04	8.60	5.56	7.75	7.55	3.45
Gemma-7B Instruct	47.79	93.25	55.96	21.98	15.51	7.99	8.45	11.32	17.24
CodeGemma-7B	46.09	91.33	53.76	21.70	13.67	6.94	5.63	8.49	3.45
CodeGemma-7B Instruct	53.49	93.48	69.57	26.22	19.82	12.50	12.68	11.32	10.34
Mistral 7B	45.67	96.01	50.28	18.56	11.37	8.68	12.68	7.55	10.34
OpenMathMistral	38.25	90.94	34.40	14.22	8.76	5.90	4.93	5.66	20.69
LLama 7B	28.67	97.93	0.71	10.53	2.92	3.82	3.52	0.94	3.45
Float 7B Instruct	27.55	89.33	1.56	12.37	5.07	5.21	2.82	4.72	13.79

Table 6: Entity Tracking performance of models categorized by the number of operations affecting the entity of interest. The count of test set instances with the number of operations affecting an entity is indicated in parentheses below the corresponding column title.